* Matrix Multiplication:-

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$mA = \begin{bmatrix} ma_{11} & ma_{12} \\ ma_{21} & ma_{22} \end{bmatrix} \qquad m \neq 0$$

If A and B are two matrices, then,

$$(A+B)^2 = A^2 + AB + BA + B^2$$

Multiplication of two matrices:-

If, $A = (a_{ij})_{m\times n}$ $\qquad 1 \le i \le m$

$B = (b_{jk})_{n\times p}$ $\qquad 1 \le j \le n$

$\qquad 1 \le k \le p$

then, $AB = (c_{ik})_{m\times p}$

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}_{2\times 3}$$

$$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

$$AB = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

* Commutative law:-

* If $AB = BA$, then the matrices are called commutative matrices.

  Then, $(A+B)^2 = A^2 + 2AB + B^2$

* If $AB = -BA$, the the matrices are called anti commutative matrices.

  Then, $(A+B)^2 = A^2 + B^2$

* If $A = (a_{ij})_{m\times n}$ and $B = (b_{ji})_{n\times m}$ and $b_{ji} = a_{ij}$ then they are called transpose of each other.

$$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{pmatrix}$$

B = transepose of (A)

$= A' \; (A^T)$

* $(A')' = A$
* $(A+B)' = A' + B'$
* $(AB)' = B'A'$

* If $A' = A$, then A is called symmetric matrix.

Ex:- $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

* If $A' = -A$, the A is called anti-or skew-symmetric matrix.

* Every square matrix can be uniquely expressed as a sum of symmetric and skew-symmetric parts.

Prove:-

Let, A is a square matrix

$$A = \underbrace{\tfrac{1}{2}(A+A')}_{\text{Symmetric}} + \underbrace{\tfrac{1}{2}(A-A')}_{\text{skew Symmetric}}$$

$$\left(\frac{1}{2}(A+A')\right)' = \frac{1}{2}(A+A')'$$
$$= \frac{1}{2}(A' + (A')')$$
$$= \frac{1}{2}(A+A')$$

$\therefore \frac{1}{2}(A+A') =$ symmetric

And, $\left(\frac{1}{2}(A-A')\right)' = \frac{1}{2}(A-A')'$
$$= \frac{1}{2}(A' - (A')')$$
$$= \frac{1}{2}(A' - A)$$
$$= -\frac{1}{2}(A - A') = \text{Antisymmetric}$$

Let, $A = A_1 + A_2$ where, A_1 = symmetric
A_2 = skew-symmetric

$\Rightarrow A' = A_1' + A_2'$

$\Rightarrow A' = A_1 - A_2$

$\therefore A_1 = \frac{1}{2}(A + A')$ → null for a skew-symmetric square matrix

$A_2 = \frac{1}{2}(A - A')$ → null for a symmetric square matrix

which is the unique way to express.

Example:-

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

$$A + A' = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$$

$$\frac{1}{2}(A + A') = \frac{1}{2}\begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$$

$$A - A' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\therefore \frac{1}{2}(A - A') = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

M

$x^2 \geq 0$

$x \to$ real

$y^2 \leq 0$

$y \to$ imaginary

* If A is a matrix, the $\bar{A}$ is a conjugate matrix and $(\bar{A})'$ is a transpose conjugate matrix.

* If $(\bar{A})' = \overline{(A')} = A$, then $A \to$ Hermitian matrix.

* If $(\bar{A})' = -A \to$ Skew Hermitian.

* $$A = \underbrace{\frac{1}{2}(A + \bar{A}')}_{A_1} + \underbrace{\frac{1}{2}(A - \bar{A}')}_{A_2}$$

$$\bar{A_1}' = \left(\overline{\frac{1}{2}(A + \bar{A}')}\right)'$$

$$= \frac{1}{2}(\bar{A} + (\bar{\bar{A}}'))'$$

$$= \frac{1}{2}(\bar{A} + A')'$$

$$= \frac{1}{2}(\bar{A}' + A)$$

$$= \frac{1}{2}(A + \bar{A}')$$

similiarly for A_2.

* Hermitian matrix :-

for diagonal,

$a_{ii} = \alpha + i\beta$

$\overline{a_{ii}} = \alpha - i\beta$

$\therefore \alpha - i\beta = \alpha + i\beta$

$\Rightarrow 2i\beta = 0$

$\Rightarrow \beta = 0$

$\therefore$ The elements must be real or zero along the diagonal.

for skew...,

$a_{ii} = -\overline{a_{ii}}$

$\alpha + i\beta = -\alpha + i\beta$

$\Rightarrow 2\alpha = 0$

$\therefore \alpha = 0$

$\therefore$ elements must be imaginary or zero along the diagonal.

* $A = \begin{pmatrix} 0 & \alpha + i\beta \\ \alpha - i\beta & 0 \end{pmatrix}$

$\bar{A} = \begin{pmatrix} 0 & \alpha - i\beta \\ \alpha + i\beta) & 0 \end{pmatrix}$

$\therefore (\bar{A})' = \begin{pmatrix} 0 & \alpha + i\beta \\ \alpha - \beta & 0 \end{pmatrix} = A$

$\therefore$ A is a harmitian matrix.

* If a matrix is harmitian, then its skew-harmitian part is zero and converse.

* **Determinant:-**

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Co-factor of $a_{11} = (-1)^2 \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

" " $a_{12} = (-1)^3 \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

" " $a_{13} = (-1)^4 \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$a_{11}\left(\text{Co-factor of } a_{11}\right) + a_{12}\left(\text{co-factor of } a_{12}\right) + a_{13}\left(\text{Co-factor of } a_{13}\right) = D$

$a_{21}\left(\quad " \quad\right) + a_{22}\left(\quad " \quad\right) + a_{23}\left(" \right) = 0$

$a_{31}\left(\quad " \quad\right) + a_{32}\left(" \right) + a_{33}\left(" \right) = 0$

Co-factor matrix:

$A = (a_{ij})_{n \times n}$

the c is a co-factor matrix of A if,

$c = (c_{ij})_{n \times n}$ and c_{ij} = co-factor of a_{ij}

$Adj(A) = c'$ = transpose of Co-factor matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$C = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

c_{11} = Co-factor of $a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

c_{33} = " " $a_{33} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$

* $A = \begin{pmatrix} 1 & 5 \\ 3 & 7 \end{pmatrix}$

$c = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 1 \end{pmatrix}$ $\therefore adj(A) = \begin{pmatrix} 7 & -5 \\ -3 & 1 \end{pmatrix}$

c_{11} = co-factor of $a_{11} = 7$

c_{12} = " " $a_{12} = -3$

" " $a_{21} = -5$

c_{21} = " " $a_{22} = 1$

c_{22} " "

* $A^{-1} = \dfrac{Adj(A)}{|A|}$

$= -\frac{1}{8}\begin{pmatrix} 7 & -5 \\ -3 & 1 \end{pmatrix}$

$\therefore A \cdot A^{-1} = \left(-\frac{1}{8}\right)\begin{pmatrix} 1 & 5 \\ 3 & 7 \end{pmatrix}\begin{pmatrix} 7 & -5 \\ -3 & 1 \end{pmatrix}$

$= -\frac{1}{8}\begin{pmatrix} -8 & 0 \\ 0 & -8 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\because AB = BA = I$, then,

$A = B^{-1}$

or $B = A^{-1}$

* $A = \begin{pmatrix} 1 & 5 \\ 3 & 7 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\sim -3\begin{pmatrix} 1 & 5 \\ 3 & 7 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\sim -\frac{1}{8}\begin{pmatrix} 1 & 5 \\ 0 & -8 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 3/8 & -1/8 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -7/8 & \frac{5}{8} \\ \frac{3}{8} & -1/8 \end{pmatrix}$

Matrix:-

$A(\text{adj } A) = \text{adj}(A)A = |A| I$.

Prove:

$$\text{Adj}(A) = c' = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & \vdots \\ \vdots & \vdots & & c_{nn} \\ c_{n1} & c_{n2} & \cdots & \end{bmatrix}$$

$$= \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \vdots & \vdots & & \vdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$A(\text{adj}(A)) = \begin{bmatrix} a_{11}c_{11} + a_{12}c_{12} + \cdots + a_{nn}c_{1n} & \\ a_{21}c_{11} + a_{22}c_{12} + \cdots\cdots a_{2n}c_{1n} & \boxed{} \\ \vdots & \\ a_{n1}c_{11} + \cdots\cdots a_{nn}c_{1n} & \boxed{} \end{bmatrix}$$

$$\begin{matrix} a_{21}a_{21} + a_{22}\cancel{c}_{22} + \cdots + a_{2n}c_{2n} \\ \vdots \\ a_{n1}c_{n1} + a_{n2}c_{n2} + \cdots + a_{nn}c_{nn} \end{matrix} \Bigg]$$

$$= \begin{bmatrix} |A| & 0 & \cdots & 0 \\ 0 & |A| & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & & \cdots & |A| \end{bmatrix} = |A| I \text{ (proved)}.$$

Now, $A(\operatorname{adj}(A)) = |A| I$

$\Rightarrow A^{-1} A(\operatorname{adj} A) = |A| A^{-1} I$

$= |A| A^{-1}$

$\Rightarrow I(\operatorname{adj}(A)) = |A| A^{-1}$

$\Rightarrow A^{-1} = \dfrac{\operatorname{adj}(A)}{|A|}$ $[|A| \neq 0]$

$$A = \begin{bmatrix} 4 & 6 \\ 5 & 3 \end{bmatrix}$$

c_{11} = co factor of $a_{11} = 3$

c_{12} = " " " $a_{12} = -5$

c_{21} = " " " $a_{21} = -6$

c_{22} = " " " $a_{22} = 4$

$$\therefore C = \begin{bmatrix} 3 & -5 \\ -6 & 4 \end{bmatrix}$$

$\therefore \text{Adj}(A) = C' = \begin{bmatrix} 3 & -6 \\ -5 & 9 \end{bmatrix}$

$|A| = -18$

$$A(\text{adj}(A)) = \begin{bmatrix} 4 & 6 \\ 5 & 3 \end{bmatrix}\begin{bmatrix} 3 & -6 \\ -5 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -18 & 0 \\ 0 & -18 \end{bmatrix}$$

$$= -18\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= |A| \, I_2$$

* Another way to determine inverse:- (Elementary transformation)

$$A \left[\begin{array}{cc|cc} 4 & 6 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array}\right]$$

$$\approx \left[\begin{array}{cc|cc} 1 & -3 & -1 & 1 \\ 5 & 3 & 0 & 1 \end{array}\right] \quad R_1 = R_2 - 1$$

$$\approx \left[\begin{array}{cc|cc} 1 & -3 & -1 & 1 \\ 0 & -18 & 5 & -4 \end{array}\right]$$

$$\sim \left[\begin{array}{cc|cc} 1 & -3 & -1 & 1 \\ 0 & 1 & \frac{5}{18} & -\frac{4}{18} \end{array}\right]$$

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -\frac{3}{18} & \frac{6}{18} \\ 0 & 1 & \frac{5}{18} & -\frac{4}{18} \end{array}\right]$$

* Rank :-

$A = (a_{ij})_{m \times n}$

minors → determinant of square sub matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix}$$

$$\left.\begin{array}{cc} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ \begin{vmatrix} a_{11} & a_{14} \\ a_{21} & a_{24} \end{vmatrix} & \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ \begin{vmatrix} a_{12} & a_{14} \\ a_{22} & a_{24} \end{vmatrix} & \begin{vmatrix} a_{13} & a_{14} \\ a_{23} & a_{24} \end{vmatrix} \end{array}\right\} \text{minors}$$

If

(i) all the minors of $(r+1)$ order will be zero "

(ii) Atleast one minor of order r will not zero.

then,

$\rho(A) = r$
↘ Rank of A

Example:-

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

here minors of $(r+1) = 1+1 = 2$ is 0.

But minors of $r = 1$ is not equal to zero.

$\therefore \rho(A) = 1$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 9 \end{bmatrix}$$

for 2×2 by orders, not all minors will be zero and we will not get minors for three or higher orders.

$\therefore \rho(A) = 2.$

CamScanner

$$\left.\begin{aligned} x+y&=1\\ 2x+2y&=1\end{aligned}\right\}$$

$Ax = B$

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$; $x = \begin{pmatrix} x \\ y \end{pmatrix}$. $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 1 \end{array}\right] \longrightarrow$ rank 1 = 2

$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \longrightarrow$ rank2 = 1 $\therefore$ rank1 $\neq$ rank 2

2×2

$\therefore$ No solution of the equations.

If ranks are equal for the above two matrix and numbers of variables are equal to the rank, then there will exist a single solution of the equation.

If ranks are not equal, then there is no solution of the equation.

Matrix:-

* Application of rank in Linear System:-

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{12}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}\right\}$$

$AX = B$ where, $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$

Augmented Matrix;-

$A^* = [A : B]$ $\quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$ $\quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

If, $\rho(A^*) = \rho(A)$, then consistent and solution is available for the equations.

If, $\rho(A^*) \neq \rho(A)$, then inconsistent.

Example:-

$$x + y = 1$$
$$x - y = 0$$

$$AX = B$$

$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$

* $A^* = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ rank = 2

$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ rank = 2

$\therefore$ consistent.

for, $x+y=1$
$x+y=2$

$\because A^* = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ rank = 2

$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$ . . . . (i)

rank of $A = 1$

$\therefore$ not consistent.

consistent → Unique Solution
$\rho(A^*) = \rho(A) =$ number of variables

→ Many Solutions
$\rho(A^*) = \rho(A) <$ numbers of variables

Rank:-

If, A is matrix,

$A \sim \begin{cases} [I_r] \\ [I_r \; 0] \\ \begin{bmatrix} I_r \\ 0 \end{bmatrix} \\ \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \end{cases}$

I_r = Unit matrix of order r

Echolon Form:- → No column operation (Restriction)

$A \sim$ [Upper tangular form]

non zero row = all the elements are zero.

$\rho(A) =$ number of non zero row

from (i) ⇒

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \sim [I_2 \; 0]$$

Now, $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}$

Rank :-

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 7 & 9 & 11 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -1 & -2 & -3 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \cdots (i)$$

$\therefore \rho(A) = 2$ = Number of non-zero rows.

(i) ⇒

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -3 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Example:

$x+y=1$

$2x+2y=2$

$A^* : [A \; B]$

$\therefore A^* = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore \rho(A^*) = 1$

$\therefore \rho(A) = 1$

$\therefore$ Infinite many solutions are available.

$x + by = 1$

$bx + y = 1$

$A = \begin{bmatrix} 1 & b \\ b & 1 \end{bmatrix}; \; B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$A^* = \begin{bmatrix} 1 & b & 1 \\ b & 1 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & b & 1 \\ 0 & 1-b^2 & 1-b \end{bmatrix}$

$\sim \begin{bmatrix} 1 & b & 1 \\ 0 & (1+b)(1-b) & (1-b) \end{bmatrix}$

$\therefore$ If $(b \neq 1)$, then $\rho(A^*) = 2 \rightarrow$ many solution.

If $b = -1$, $\rho(A) = 1$ $\}$ no solution

If $b \neq 1$ and $b \neq -1$, the an unique solution is available.

If $AX = B$ is a linear system, There is a trivial solution.

If $B = 0 \longmapsto$ Homogenous equ.

$B \neq [0] \longrightarrow$ Non-Homogenous equ.

$$a_{11}x + a_{12}y = 0$$
$$a_{21}x + a_{22}y = 0$$

Example:

(i) $x + y = 0$

$x - y = 0$

rank, $\rho(A^*) = \rho(A) = 2$

$\therefore$ unique solution and solution is $(0,0)$.

(ii) $x + y = 0$

$x + y = 0$

rank, $\rho(A^*) = \rho(A) = 1$

$\therefore$ many solutions available.